VIDYAPEETH

BATCH CODE: 12- AN181MA

SUBJECT NAME: Zoology

CHAPTER NAME:
Structural organization in animal.

Lecture No. 09

By- Murshid Zafar Sir

Today's Goal

Subtopic .. C. Tissue

Subtopic ..

Subtopic ..

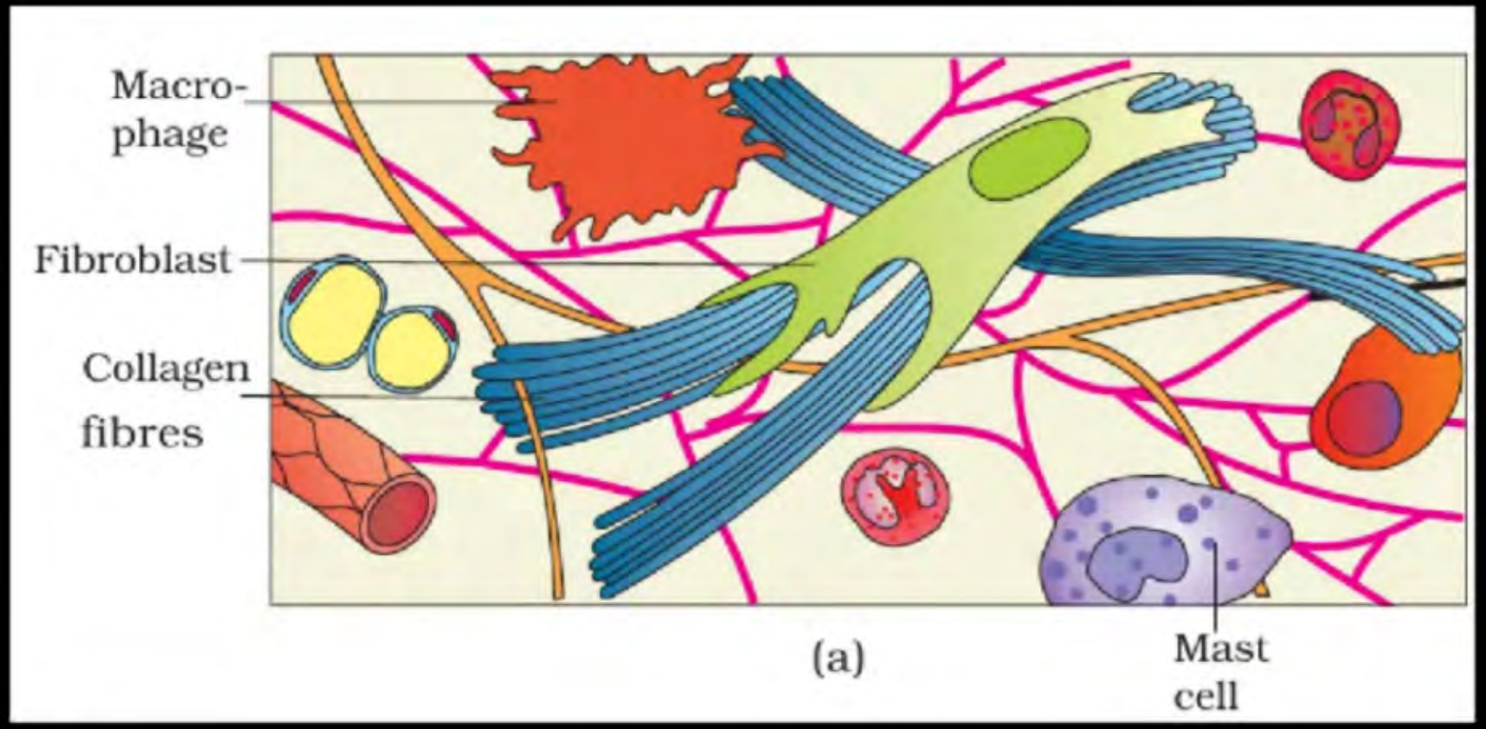
Types of Connective Tissue

Three type.

- **Loose C.T**
  - Matrix ↑ (soft)
  - Fibre ↓
  - 2 Type
    - **Areolar C.T**
    - **Adipose C.T**

- **Dense C.T**
  - Matrix ↓ (soft)
  - Fibre ↑
  - 2 Type
    - **Dense regular C.T**
      - → orientation of fibre = Regular / Parallel
    - **Dense irregular C.T**
      - → orientation of Fibre = Irregular

- **Specialised C.T**
  - **Skeletal C.T**
    - Matrix = Mineralised
    - Bone
    - Cartilage
  - **Fluid C.T**
    - Blood
    - Lymph
    - Matrix = Fluid



(a)

Loose C.T

- Matrix ↑
- Matrix Soft
- Fibre ↓
- Two Type

Areolar C.T

- Cell
  - Fibroblast
  - Macrophage
  - Mast cell.
- Matrix ↑
- Fibre ↓

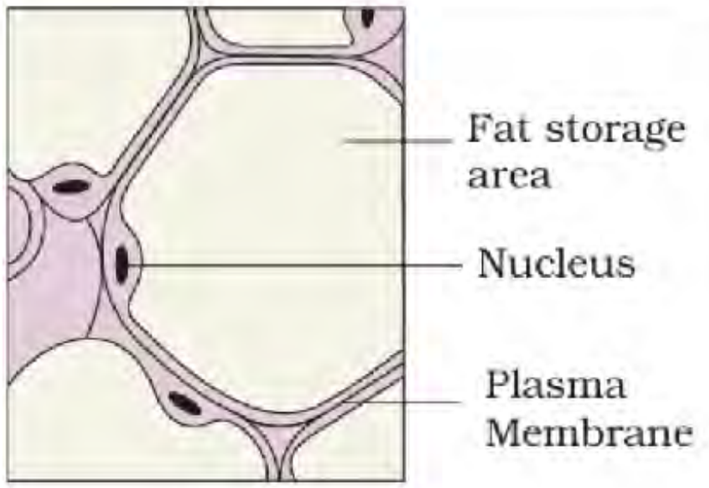
exp:- Beneath the skin
Tela subcutanea (NCERT) X.

Adipose C.T

- Cell
  - Fibroblast
  - Macrophage
  - Mast cell
  - Adipocyte (अधिक) → Store fat.
- Matrix ↑
- Fibre ↓

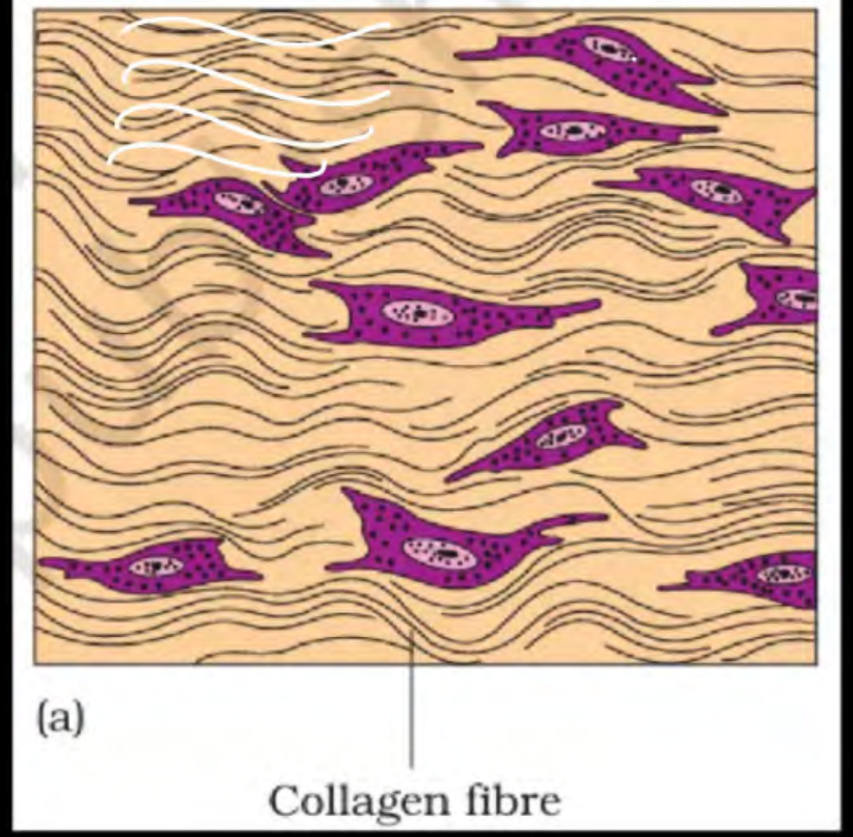
Exp:- present Beneath the skin

Adipose C.T — acts as
- Shock absorber
- Insulator.



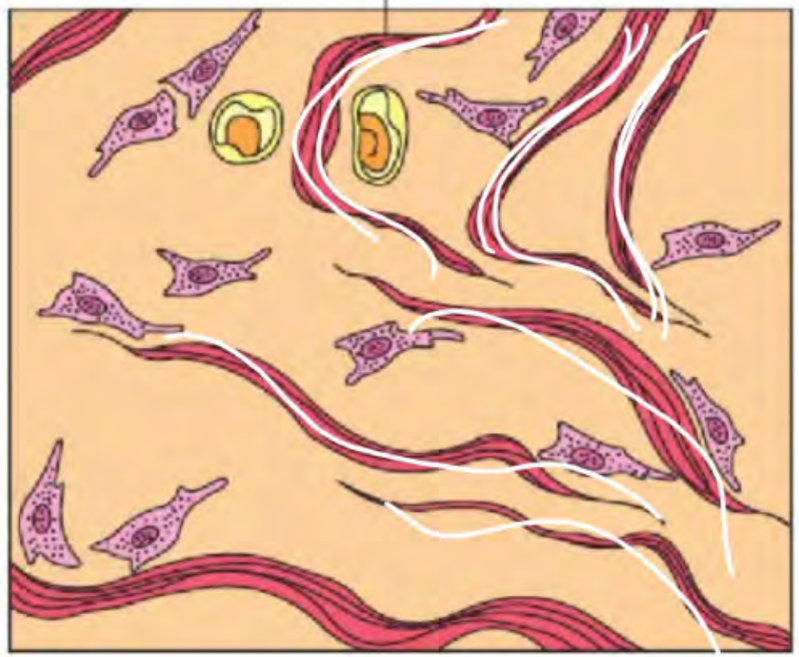
(b)

include cartilage, bone, adipose, and blood. In all connective tissues except blood, the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissue. These cells also secrete modified polysaccharides, which accumulate between cells and fibres and act as matrix (ground substance). Connective tissues are classified into three types: (i) Loose connective tissue, (ii) Dense connective tissue and (iii) Specialised connective tissue.


(a)
Collagen fibre

Loose connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance, for example, **areolar tissue** present beneath the skin (Figure 7.4). Often it serves as a support framework for epithelium. It contains fibroblasts (cells that produce and secrete fibres), macrophages and mast cells. **Adipose tissue** is another type of loose connective tissue located mainly beneath the skin. The cells of this tissue are specialised to store fats. The excess of nutrients which are not used immediately are converted into fats and are stored in this tissue.

Fibres and fibroblasts are compactly packed in the **dense connective tissues**. Orientation of fibres show a regular or irregular pattern and are called **dense regular** and **dense irregular tissues**. In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones and ligaments which attach one bone to another are examples of this tissue. Dense irregular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently (Figure 7.5). This tissue is present in the skin. Cartilage,



(b)

Figure 7.5 Dense connective tissue:
(a) Dense regular
(b) Dense irregular

bones and blood are various types of **specialised connective tissues**.

The intercellular material of **cartilage** is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them (Figure 7.6a). Most of the cartilages in vertebrate embryos are replaced by bones in adults. Cartilage is present in the tip of nose, outer ear joints, between adjacent bones of the vertebral column, limbs and hands in adults.

Topic : Connective Tissue

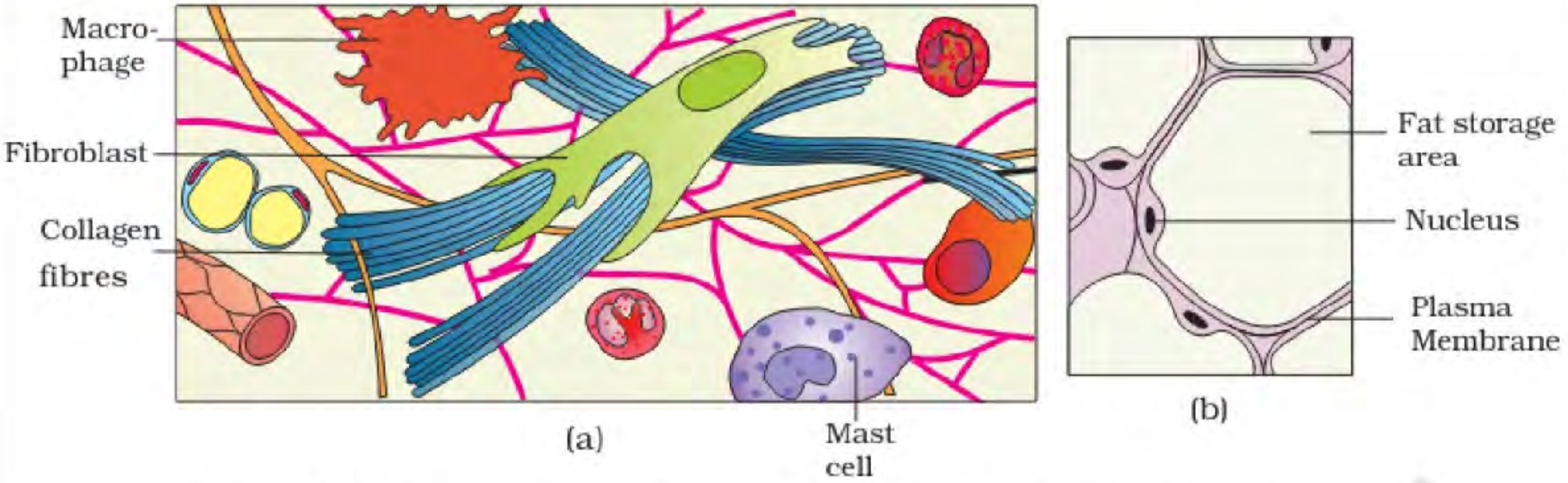


Figure 7.4 Loose connective tissue : (a) Areolar tissue (b) Adipose tissue

Topic : loose Connective Tissue

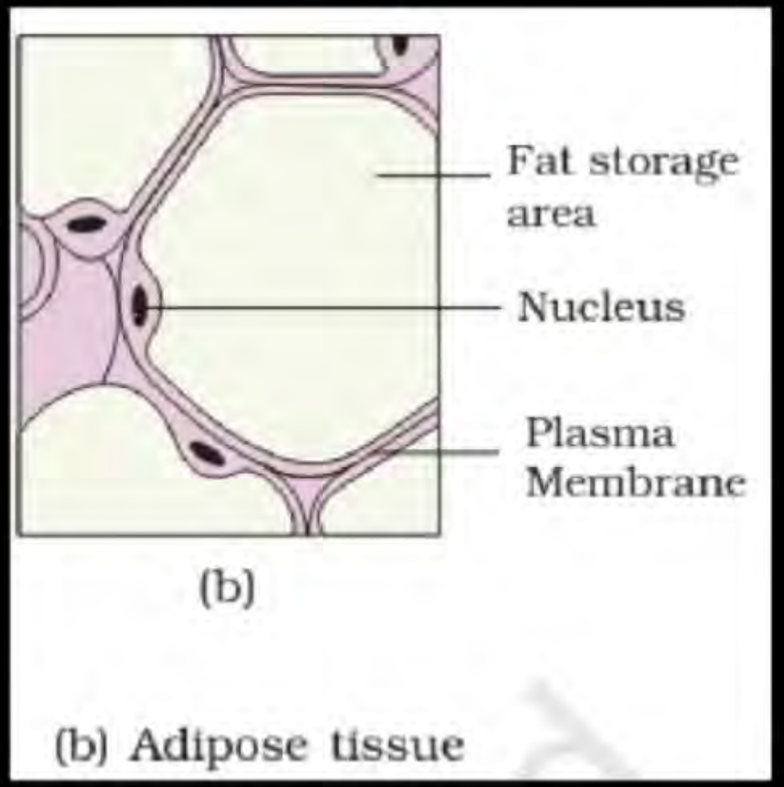
- **(A) Areolar tissue** present beneath the skin .
- It serves as a support framework for epithelium.
- It contains fibroblasts (cells that produce and secrete fibres), macrophages and mast cells.

Out of NCERT

PYQ = Areolar C-T Joins Integument with muscle.

- **(B) Adipose tissue** is another type of loose connective tissue located mainly beneath the skin.

- The cells of this tissue are specialised to store fats. The excess of nutrients which are not used immediately are converted into fats and are stored in this tissue.



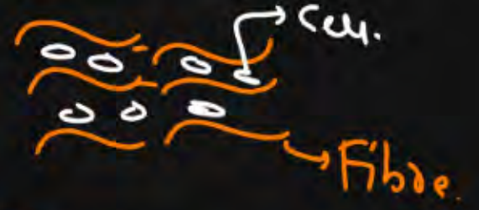
(b) Adipose tissue

BABLI

Dense C.T

Dense Regular C.T

Orientation of fibre = regular



Note:- Dense regular C.T रस्सी बनाता है
↳ Jodne का काम करती
connect

Example: Ligament, Tendon.
(connect bone to bone) (connect bone to muscle)

Dense Irregular C.T

orientation fibre = Irregular.

Example: tnt in skin

Dense Connective Tissues

- Fibres and fibroblasts are compactly packed in the dense connective tissues.
- Orientation of fibres show a regular or irregular pattern and are called dense regular and dense irregular tissues.

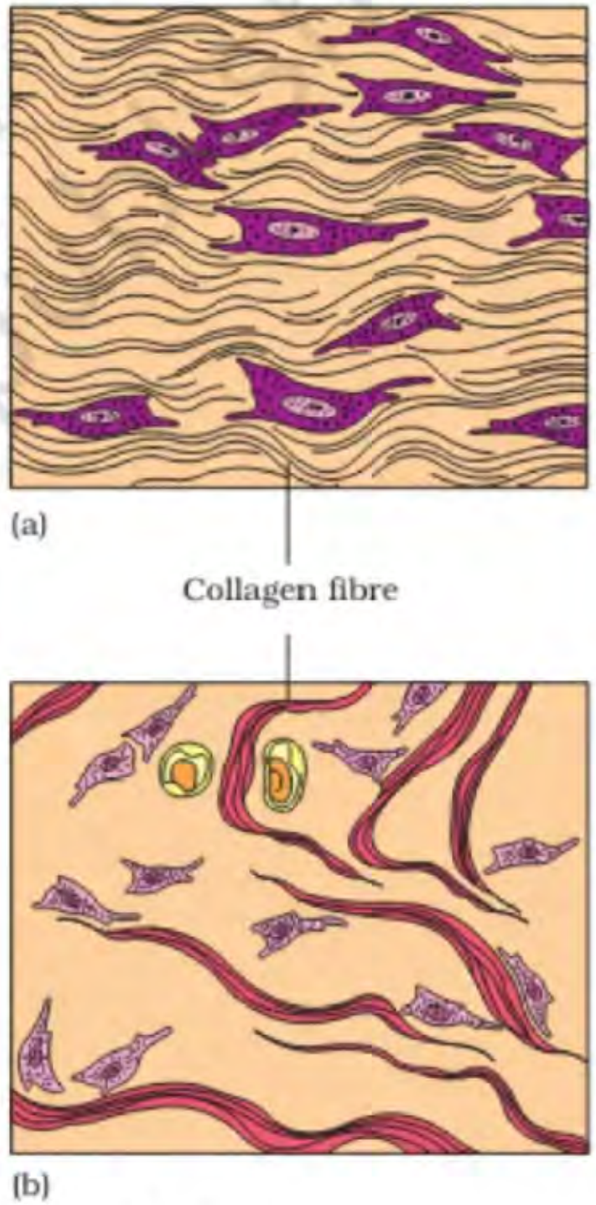
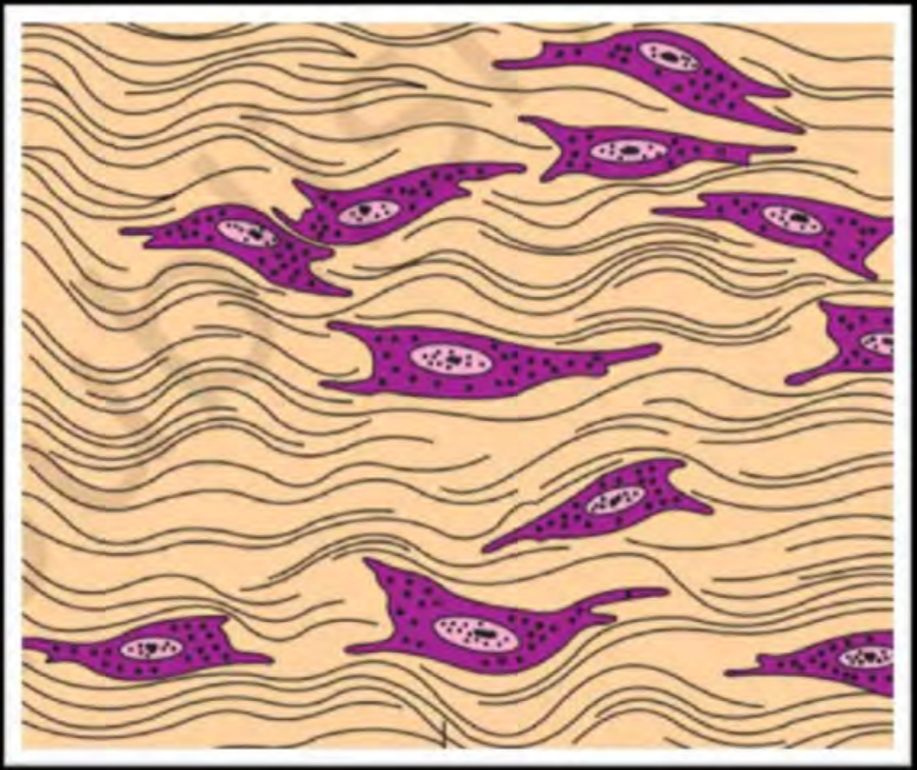


Figure 7.5 Dense connective tissue: (a) Dense regular (b) Dense irregular

(A) Dense regular :- In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres.

Example :- Tendon & ligament



(B) Dense irregular connective tissue :-

It has fibroblasts and many fibres (mostly collagen) that are oriented differently (irregular)

Example :- This tissue is present in the skin.

Specialised C.T

- Skeletal C.T
- Fluid C.T

Skeletal C.T

→ Matrix - Mineralised

Bone

Cartilage

Chondro = Cartilage
Blast = Formation
cyte = cell
clast = destroy

Cartilage

Cell

1. Chondroblast
   - → produce fibre & Matrix
   - → Cartilage forming cell.
2. → Chondrocyte
   - → produce matrix.
   - → cell of cartilage.
3. → Chondroclast
   - → Cartilage destroying cell.

Matrix

Pya
- → Solid
- → Pliable (Lachila)
- → Resist compression.

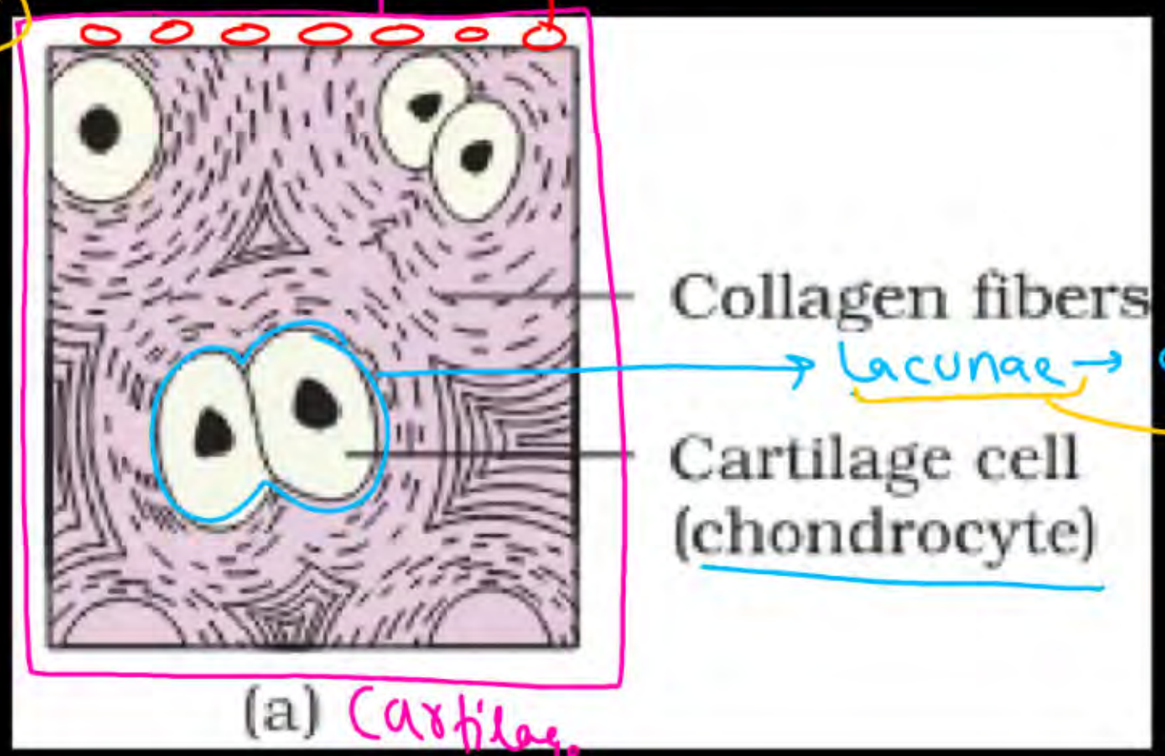
Fibre

- → Elastic fibre
- → Collagen.

Perichondrium (covering of cartilage).

Chondroblast

Avascular → C.-Tissue
→ Cartilage



(a) Cartilage

Lacunae → chondrocyte ki ccl

each lunae consist of 1 or more than 1 chondrocyte

Question

#Q. Pseudostratified ciliated columnar epithelia is observed in;

A. Respiratory tract

B. PCT

C. Urethra

D. Gastro-intestinal tract

Question

#Q. The large amoeboid cells found in areolar tissue and are also part of our innate immune system are known as;

A. Mast cells

B. Macrophages

C. Adipocytes

D. Fibroblasts

Question

#Q. **Statement-I**: Epithelial tissues are specialized for linking and connecting other tissues/organs.

Statement-II: Tendon and ligament are the types of dense irregular connective tissues.

A. Both Statement-I and Statement-II are correct.

B. Both Statement-I and Statement-II are incorrect.

C. Statement-I is correct & Statement-II is incorrect.

D. Statement-I is incorrect & Statement-II is correct.

Question

#Q. **Statement-I:** Muscle is an example of connective tissue.

Statement-II: Blood is an example of connective tissue:

A. Both Statement-I and Statement-II are correct.

B. Both Statement-I and Statement-II are incorrect.

C. Statement-I is correct & Statement-II is incorrect.

D. Statement-I is incorrect & Statement-II is correct.

Question

#Q. **Assertion (A):** Tendons attach one bone to another bone.

Reason (R): Ligaments attach bone to bone.

A. Assertion (A) is true, Reason (R) is true; Reason (R) is a correct explanation for Assertion (A).

B. Assertion (A) is true, Reason (R) is true; Reason (R) is not a correct explanation for Assertion (A).

C. Assertion (A) is true, Reason (R) is false.

D. Assertion (A) is false, Reason (R) is true.

Question

#Q. Which of the following tissue exerts the greatest control over the body's responsiveness to changing conditions?

A. Epithelial tissue

B. Connective tissue

C. Muscular tissue

D. Neural tissue

Question

#Q. The cells present in the connective tissue are responsible for,

A. Secretion of fibres

B. Secretion of ground substance or matrix

C. Both (1) and (2)

D. None of these

Question

#Q. Out of the given tissues, how many are types of connective tissue: Adipose, bones, skeletal, dense irregular, simple squamous, areolar.

C.T

A. Two

B. Four

C. Six

D. Five

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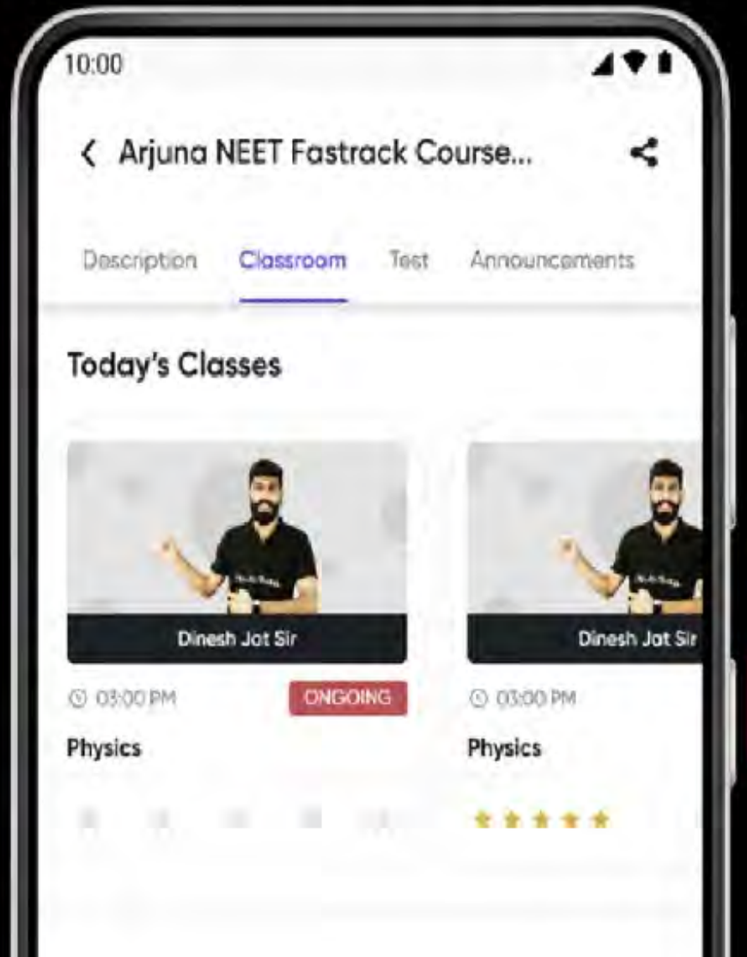

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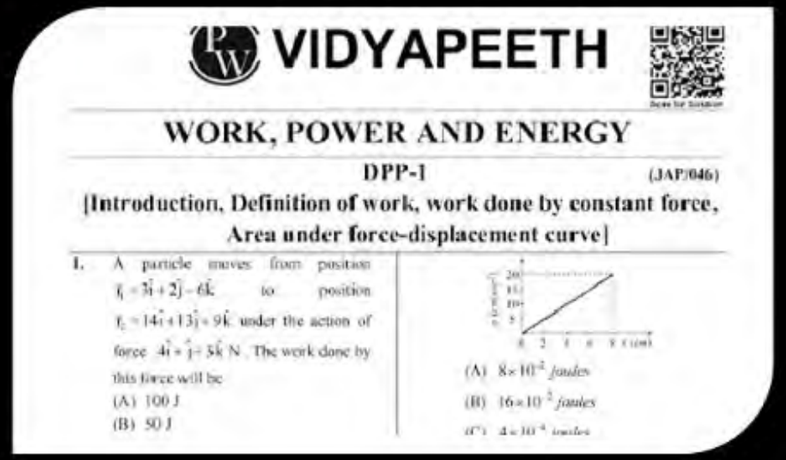
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WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be
(A) 100 J
(B) 50 J

(A) 8×10^{-2} joules
(B) 16×10^{-2} joules

Thank You!!!!

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